

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

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Step 3: Install OpenSSL

OpenSSL is used to create and manage SSL/TLS certificates.

```
sudo apt install openssl -y
```

Check the version:

```
shubham@SHUBHAM:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssl is already the newest version (3.0.13-0ubuntu3.12).
openssl set to manually installed.
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
shubham@SHUBHAM:~$
```

openssl version

Step 4: Enable SSL Module in Apache

```
sudo a2enmod ssl
```

Restart Apache:

```
sudo systemctl restart apache2
```

```
shubham@SHUBHAM:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
shubham@SHUBHAM:~$
```

Step 5: Create a Directory for Certificates

```
sudo mkdir -p /etc/ssl/mycert
```

Move to the directory:

```
cd /etc/ssl/mycert
```

```
shubham@SHUBHAM:~$ sudo systemctl restart apache2
shubham@SHUBHAM:~$ sudo mkdir -p /etc/ssl/mycert
shubham@SHUBHAM:~$ cd /etc/ssl/mycert
shubham@SHUBHAM:/etc/ssl/mycert$
```

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Step 6: Generate a Private Key

Create a 2048-bit RSA private key:

```
sudo openssl genrsa -out server.key 2048
```

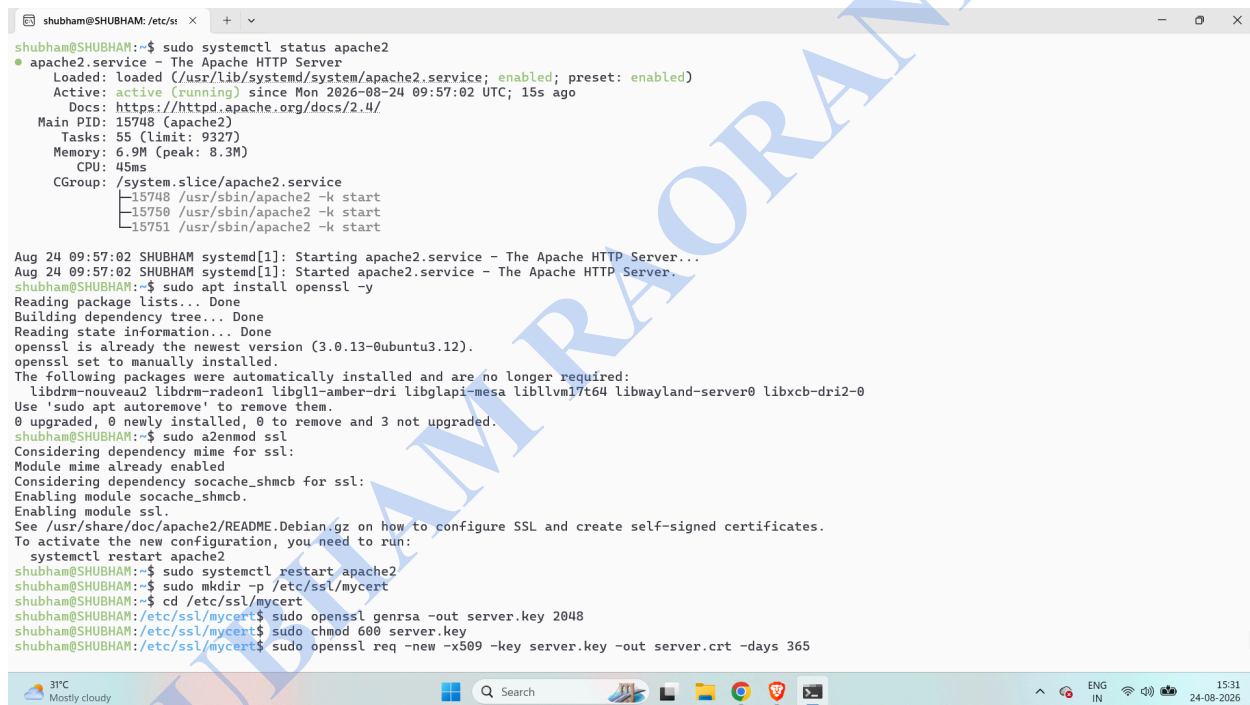
The file:

server.key

is the private key and must be kept secure.

Set proper permissions:

```
sudo chmod 600 server.key
```



```
shubham@SHUBHAM: /etc/sx
shubham@SHUBHAM:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 09:57:02 UTC; 15s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 15748 (apache2)
    Tasks: 55 (limit: 9327)
   Memory: 6.9M (peak: 8.3M)
      CPU: 45ms
   CGroup: /system.slice/apache2.service
           └─15748 /usr/sbin/apache2 -k start
             └─15750 /usr/sbin/apache2 -k start
               └─15751 /usr/sbin/apache2 -k start

Aug 24 09:57:02 SHUBHAM systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 24 09:57:02 SHUBHAM systemd[1]: Started apache2.service - The Apache HTTP Server.
shubham@SHUBHAM:~$ sudo apt install openssl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssl is already the newest version (3.0.13-0ubuntu3.12).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 3 not upgraded.
shubham@SHUBHAM:~$ sudo a2enmod ssl
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
shubham@SHUBHAM:~$ sudo systemctl restart apache2
shubham@SHUBHAM:~$ sudo mkdir -p /etc/ssl/mycert
shubham@SHUBHAM:~$ cd /etc/ssl/mycert
shubham@SHUBHAM:/etc/ssl/mycert$ sudo openssl genrsa -out server.key 2048
shubham@SHUBHAM:/etc/ssl/mycert$ sudo chmod 600 server.key
shubham@SHUBHAM:/etc/ssl/mycert$ sudo openssl req -new -x509 -key server.key -out server.crt -days 365
```

Step 7: Generate a Self-Signed SSL Certificate

Run:

```
sudo openssl req -new -x509 -key server.key -out server.crt -days 365
```

You will be asked for information such as:

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Country Name: IN
State: Maharashtra
Locality: Mumbai
Organization: My College
Organizational Unit: IT Department
Common Name: localhost
Email Address: your-email@example.com

This creates:

server.crt

The certificate will be valid for 365 days.

```
shubham@SHUBHAM:/etc/ssl/mycert$ sudo openssl req -new -x509 -key server.key -out server.crt -days 365
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:IN
State or Province Name (full name) [Some-State]:Maharashtra
Locality Name (eg, city) []:Mumbai
Organization Name (eg, company) [Internet Widgits Pty Ltd]:My College
Organizational Unit Name (eg, section) []:CS Department
Common Name (e.g. server FQDN or YOUR name) []:localhost
Email Address []:your-email@example.com
shubham@SHUBHAM:/etc/ssl/mycert$
```

Step 8: Configure Apache for HTTPS

Enable the default SSL site:

`sudo a2ensite default-ssl`

```
shubham@SHUBHAM:/etc/ssl/mycert$ sudo a2ensite default-ssl
Enabling site default-ssl.
To activate the new configuration, you need to run:
  systemctl reload apache2
shubham@SHUBHAM:/etc/ssl/mycert$
```

Edit the SSL configuration:

`sudo nano /etc/apache2/sites-available/default-ssl.conf`

Find these lines:

SSLCertificateFile /etc/ssl/certs/ssl-cert-snakeoil.pem
SSLCertificateKeyFile /etc/ssl/private/ssl-cert-snakeoil.key

Replace them with:

SSLCertificateFile /etc/ssl/mycert/server.crt

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SSLCertificateKeyFile /etc/ssl/mycert/server.key

```
shubham@SHUBHAM: /etc/ssl *
GNU nano 7.2 /etc/apache2/sites-available/default-ssl.conf *
<VirtualHost *:443>
    ServerAdmin webmaster@localhost

    DocumentRoot /var/www/html

    # Available loglevels: trace8, ..., trace1, debug, info, notice, warn,
    # error, crit, alert, emerg.
    # It is also possible to configure the loglevel for particular
    # modules, e.g.
    #LogLevel info ssl:warn

    ErrorLog ${APACHE_LOG_DIR}/error.log
    CustomLog ${APACHE_LOG_DIR}/access.log combined

    # For most configuration files from conf-available/, which are
    # enabled or disabled at a global level, it is possible to
    # include a line for only one particular virtual host. For example the
    # following line enables the CGI configuration for this host only
    # after it has been globally disabled with "a2disconf".
    #Include conf-available/serve-cgi-bin.conf

    #
    # SSL Engine Switch:
    # Enable/Disable SSL for this virtual host.
    #
    SSLEngine on

    #
    # A self-signed (snakeoil) certificate can be created by installing
    # the ssl-cert package. See
    # /usr/share/doc/apache2/README.Debian.gz for more info.
    # If both key and certificate are stored in the same file, only the
    # SSLCertificateFile directive is needed.
    SSLCertificateFile /etc/ssl/mycert/server.crt
    SSLCertificateKeyFile /etc/ssl/mycert/server.key

    #
    # Server Certificate Chain:
    # Point SSLCertificateChainFile at a file containing the
    # concatenation of PEM encoded CA certificates which form the
    # certificate chain for the server certificate. Alternatively
```

Step 9: Test Apache Configuration

Run: `sudo apache2ctl configtest`

Expected result:

Syntax OK

```
shubham@SHUBHAM: /etc/ssl/mycert$ sudo nano /etc/apache2/sites-available/default-ssl.conf
shubham@SHUBHAM: /etc/ssl/mycert$ sudo apache2ctl configtest
Syntax OK
shubham@SHUBHAM: /etc/ssl/mycert$
```

Step 10: Restart Apache

`sudo systemctl restart apache2`

Check the service: `sudo systemctl status apache2`

```
shubham@SHUBHAM: /etc/ssl/mycert$ sudo systemctl restart apache2
shubham@SHUBHAM: /etc/ssl/mycert$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
  Active: active (running) since Mon 2026-08-24 10:14:03 UTC; 6s ago
  Docs: https://httpd.apache.org/docs/2.4/
  Process: 16134 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 16137 (apache2)
  Tasks: 55 (limit: 9327)
  Memory: 7.6M (peak: 7.9M)
  CPU: 47ms
  CGroup: /system.slice/apache2.service
          └─16137 /usr/sbin/apache2 -k start
            └─16139 /usr/sbin/apache2 -k start
              └─16140 /usr/sbin/apache2 -k start

Aug 24 10:14:03 SHUBHAM systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 24 10:14:03 SHUBHAM systemd[1]: Started apache2.service - The Apache HTTP Server.
shubham@SHUBHAM: /etc/ssl/mycert$
```

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Step 11: Allow HTTPS Through Firewall

If UFW firewall is enabled:

```
sudo ufw allow 443/tcp
```

Or:

```
sudo ufw allow 'Apache Full'
```

Check firewall status:

```
sudo ufw status
```

```
shubham@SHUBHAM:/etc/ssl/mycert$ sudo ufw status
Status: inactive
shubham@SHUBHAM:/etc/ssl/mycert$
```



Step 12: Test Secure HTTPS Communication

Open a browser and visit:

```
https://localhost
```

You should see:

```
https://
```

with a security warning if you are using a self-signed certificate. This is normal because the certificate was not issued by a trusted Certificate Authority.

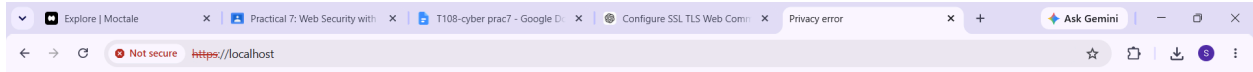
Click:

Advanced → Accept the Risk and Continue

The Apache web page should now open using HTTPS.

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Your connection is not private

Attackers might be trying to steal your information from **localhost** (for example, passwords, messages, or credit cards). [Learn more about this warning](#)

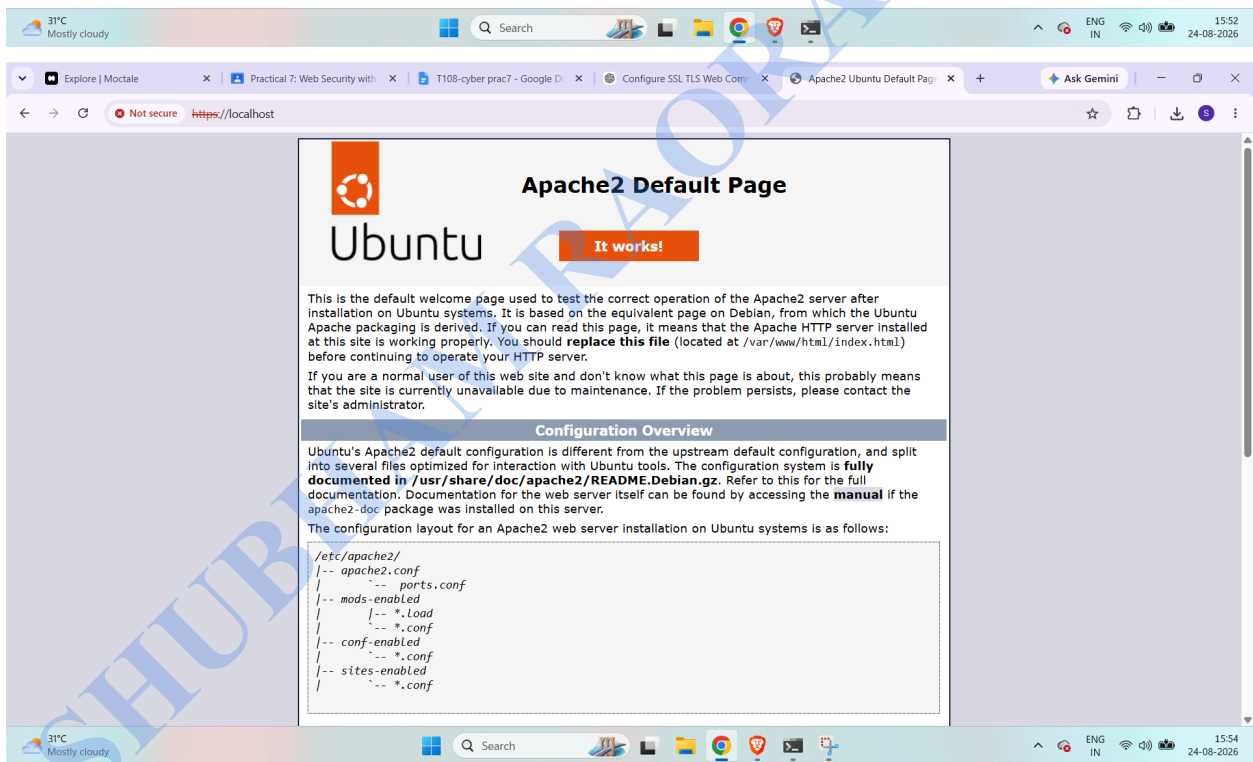
NET-ERR_CERT_AUTHORITY_INVALID



Turn on enhanced protection to get Chrome's highest level of security

Advanced

Back to safety



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Step 14: Check the Certificate

Display certificate information:

```
openssl x509 -in /etc/ssl/mycert/server.crt -text -noout
```

You can check:

- Certificate issuer
- Validity period
- Public key
- Subject/Common Name
- Signature algorithm

```
shubham@SHUBHAM: /etc/ssl$  
read R BLOCK  
closed  
shubham@SHUBHAM:/etc/ssl/mycert$ openssl x509 -in /etc/ssl/mycert/server.crt -text -noout  
Certificate:  
Data:  
  Version: 3 (0x2)  
  Serial Number:  
    3a:b5:a7:12:2c:de:b1:0f:eb:b2:c2:44:f0:b7:bf:29:96:28:3c:d0  
  Signature Algorithm: sha256WithRSAEncryption  
  Issuer: C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = your-email@example.com  
  Validity  
    Not Before: Aug 24 10:01:26 2026 GMT  
    Not After : Aug 24 10:01:26 2027 GMT  
  Subject: C = IN, ST = Maharashtra, L = Mumbai, O = My College, OU = CS Department, CN = localhost, emailAddress = your-email@example.com  
  Subject Public Key Info:  
    Public Key Algorithm: rsaEncryption  
    Public-Key: (2048 bit)  
      Modulus:  
        00:bc:3f:c2:81:b0:89:bd:9c:74:a9:0c:9c:5e:d0:  
        f5:ab:b8:d2:db:06:55:0d:86:b7:0a:18:a3:7b:50:  
        be:80:98:fa:24:1a:50:78:42:92:09:47:24:62:ab:  
        ce:d0:0c:ae:88:e0:fc:2c:32:77:65:4d:a9:59:4b:  
        76:4a:29:1d:06:69:01:cc:14:3e:62:cb:62:cf:bd:  
        06:69:06:05:0a:7b:8c:52:54:3d:b7:4f:12:bc:1e:  
        77:44:7a:c8:e8:be:a7:0e:fc:1d:19:30:20:c9:24:  
        f8:d0:81:41:39:69:b9:78:2e:5c:06:4a:05:d9:af:  
        9b:f7:ef:80:20:d6:40:11:1a:7d:43:72:ea:15:40:  
        73:54:ca:0f:0f:34:09:91:94:34:be:cl:0e:99:91:  
        ba:34:c2:d8:41:0b:ef:78:41:46:9e:ac:80:52:c4:  
        a8:78:c9:ee:0b:65:d0:93:91:b2:44:40:67:01:2c:  
        06:94:44:49:82:e4:db:cd:01:e7:0f:b6:96:5c:64:  
        72:5d:fb:f3:6e:82:c0:fa:49:73:33:b8:bb:68:4c:  
        1a:99:00:76:db:51:85:50:bb:c8:3e:e6:f2:56:cc:  
        b2:51:e2:c2:b6:98:5e:fc:bc:6c:26:f4:9a:23:6a:  
        45:c1:69:f0:72:d1:00:12:8d:32:00:e7:cf:e2:ea:  
        e8:e5  
      Exponent: 65537 (0x10001)  
  X509v3 extensions:  
    X509v3 Subject Key Identifier:  
      03:DE:76:EB:46:5E:5A:E8:23:51:FE:21:A6:A2:F6:C2:09:AA:4E:73  
    X509v3 Authority Key Identifier:  
      03:DE:76:EB:46:5E:5A:E8:23:51:FE:21:A6:A2:F6:C2:09:AA:4E:73  
  Signature Algorithm: sha256WithRSAEncryption  
  Signature Value:  
    1b:04:87:40:08:3d:fb:69:aa:dc:54:fd:29:fd:b7:3f:d3:c5:  
    58:b7:5b:ba:b6:41:b5:3b:d2:00:42:e1:81:3d:1a:46:a2:70:  
    b5:da:91:06:04:22:48:21:48:b2:8e:5b:5a:8a:d0:c3:8f:  
    d0:d0:95:b3:97:21:10:60:92:c3:05:5c:17:ee:8b:8e:fd:1f:  
    76:83:a1:bc:90:33:e6:17:4a:0e:9f:31:d5:34:3a:a4:68:b8:  
    22:5e:64:9d:a1:c0:db:22:f0:63:4b:82:50:b3:6d:fb:21:19:  
    09:d5:15:d4:e6:8e:f4:ff:d8:0c:f7:f1:be:f1:23:b4:01:38:  
    0b:2f:14:f7:1d:e4:b9:17:79:b8:22:90:71:5b:6a:82:da:29:  
    7c:57:4a:27:9d:81:7e:71:48:67:e2:3e:66:f9:15:52:a3:63:  
    99:33:48:3e:e6:2f:94:f2:b4:bc:2d:0b:61:1d:48:35:de:09:  
    2a:b6:ae:c6:f4:65:93:fc:37:16:08:ef:38:b2:d9:14:08:3a:  
    81:11:04:75:7e:f2:f4:e7:98:9f:5b:d8:e1:95:dd:18:fb:87:  
    2e:33:51:24:c3:ff:4a:26:d2:b0:a2:ce:ab:16:72:99:a5:1f:  
    82:ae:01:65:9b:2a:e2:c5:85:bd:44:e9:10:6f:65:93:b6:0b:  
    33:48:1b:15  
shubham@SHUBHAM:/etc/ssl/mycert$
```

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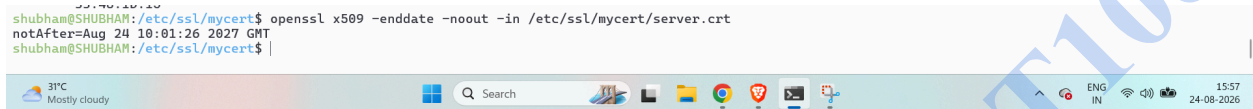
Step 15: Secure Certificate Management

Check certificate expiry date:

```
openssl x509 -enddate -noout -in /etc/ssl/mycert/server.crt
```

Example output:

```
notAfter=Aug 24 12:00:00 2027 GMT
```



Important certificate management practices:

1. Keep the private key confidential.
2. Set restricted permissions: `sudo chmod 600 server.key`
3. Monitor certificate expiry.
4. Renew certificates before expiration.
5. Use certificates issued by a trusted Certificate Authority (CA) for real websites.
6. Disable old and insecure SSL/TLS versions.

Step 16: Configure Strong TLS Security

Edit the SSL configuration:

```
sudo nano /etc/apache2/mods-available/ssl.conf
```

Ensure older protocols are disabled and only modern TLS is used, for example:

```
SSLProtocol -all +TLSv1.2 +TLSv1.3
```

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```
shubham@SHUBHAM: /etc/ss - + v
GNU nano 7.2 /etc/apache2/mods-available/ssl.conf
# List the ciphers that the client is permitted to negotiate. See the
# ciphers(1) man page from the openssl package for list of all available
# options.
# Enable only secure ciphers:
SSLCipherSuite HIGH:!aNULL

# SSL server cipher order preference:
# Use server priorities for cipher algorithm choice.
# Clients may prefer lower grade encryption. You should enable this
# option if you want to enforce stronger encryption, and can afford
# the CPU cost, and did not override SSLCipherSuite in a way that puts
# insecure ciphers first.
# Default: Off
#SSLHonorCipherOrder on

# The protocols to enable.
# Available values: all, SSLv2, TLSv1, TLSv1.1, TLSv1.2
# SSL v2 is no longer supported
SSLProtocol -all +TLSv1.2 +TLSv1.3

# Allow insecure renegotiation with clients which do not yet support the
# secure renegotiation protocol. Default: Off
#SSLInsecureRenegotiation on

# Whether to forbid non-SNI clients to access name based virtual hosts.
# Default: Off
#SSLStrictSNIVHostCheck On

# Warning: Session Tickets require regular reloading of the server!
# Make sure you do this (e.g. via logrotate) before changing this setting!
SSLSessionTickets off

[ Wrote 83 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^G Location   ^U Undo       ^M Set Mark   ^M To Bracket
^X Exit      ^R Read File  ^R Replace    ^U Paste      ^J Justify    ^_ Go To Line  ^- Redo       ^- Copy       ^_ Where Was

31°C Mostly cloudy  Search  16:00 24-08-2026
```

Restart Apache: `sudo systemctl restart apache2`

```
Syntax OK
shubham@SHUBHAM:/etc/ssl/mycert$ sudo systemctl restart apache2
shubham@SHUBHAM:/etc/ssl/mycert$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-08-24 10:31:33 UTC; 6s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 16803 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
    Main PID: 16806 (apache2)
       Tasks: 55 (limit: 9327)
      Memory: 7.1M (peak: 8.7M)
         CPU: 51ms
    CGroup: /system.slice/apache2.service
            └─16806 /usr/sbin/apache2 -k start
              └─16808 /usr/sbin/apache2 -k start
                └─16809 /usr/sbin/apache2 -k start

Aug 24 10:31:33 SHUBHAM systemd[1]: Starting apache2.service - The Apache HTTP Server...
Aug 24 10:31:33 SHUBHAM systemd[1]: Started apache2.service - The Apache HTTP Server.
shubham@SHUBHAM:/etc/ssl/mycert$
```